Speech Emotion Recognition (SER) Project Report

CS-503 Machine Learning

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1 Abstract

This report presents an utterance-level Speech Emotion Recognition (SER) pipeline using two public datasets: RAVDESS and CREMA-D. The system includes standardized preprocessing, two feature families (MFCC and pretrained wav2vec2 embeddings), three temporal pooling strategies (mean, max, mean+std), and a Logistic Regression classifier with class balancing. We evaluate both in-domain performance (RAVDESS $\rightarrow$ RAVDESS with strict speaker-independent split) and cross-domain performance (RAVDESS $\rightarrow$ CREMA-D). Results show strong in-domain performance with wav2vec2, while cross-domain generalization remains challenging, with MFCC+max giving the best cross-domain Macro-F1 in the current setup.

2 Problem Statement

Speech Emotion Recognition aims to infer emotion from speech acoustics (prosody, intensity, spectral patterns), independent of lexical content. The core challenge is generalization across speakers and datasets. The project goal is to build a modular SER pipeline and evaluate both:

- **In-domain**: train/test on the same dataset with speaker-independent split
- **Cross-domain**: train on one dataset and test on another

3 Datasets and Label Space

3.1 Datasets Used

- **RAVDESS**: used for in-domain and as training source for cross-domain
- **CREMA-D**: used as out-of-domain test dataset

3.2 Unified Label Set

To ensure compatibility across datasets, we use:

$$\{\texttt{angry, disgust, fear, happy, neutral, sad}\}$$

RAVDESS `calm` is merged into `neutral`; unsupported labels (e.g., `surprise` in cross-domain setting) are excluded where required.

4 Pipeline Design

4.1 Preprocessing

All audio is standardized via:

- Mono conversion
- Resampling to 16 kHz
- Peak normalization

4.2 Feature Extraction

- **MFCC baseline**: 13 coefficients/frame via `librosa`
- **wav2vec2**: `facebook/wav2vec2-base`, last hidden representations

4.3 Pooling Study

Variable-length frame sequences are converted to fixed-size vectors with:

- `mean`
- `max`
- `mean_std`

4.4 Classifier

Logistic Regression with:

- StandardScaler
- `class_weight = "balanced"`

4.5 Evaluation Metrics

- Accuracy
- Macro-F1
- Confusion Matrix

5 Reproducibility and Engineering Improvements

5.1 Speaker-Independent Split

RAVDESS split is deterministic with seed 42 and no speaker overlap.
Saved split (`results/split_info.json`):

- Train speakers: 19
- Test speakers: 5
- Overlap: 0

5.2 Resumable wav2vec2 Caching

To make full runs feasible:

- Per-file embedding cache in `features/ravdess` and `features/crema_d`
- Recompute only if missing (or forced)
- Safe resume after interruption

Observed cache completeness after full run:

- RAVDESS cached embeddings: 1248 files
- CREMA-D cached embeddings: 7442 files

5.3 Result Logging

All experiments are exported to `results/`:

- `results.json`, `results.csv`
- confusion matrices: `cm_{feature}_{pooling}_{eval}.csv`
- `final_comparison.csv`

6 Final Results

6.1 Wav2Vec2 Results

Pooling	In-Domain Acc	In-Domain F1	Cross-Domain Acc	Cross-Domain F1
mean	0.5731	0.5753	0.2216	0.1869
max	0.5885	0.5852	0.2025	0.1782
mean_std	0.5808	0.5836	0.2166	0.1766

6.2 MFCC Results

Pooling	In-Domain Acc	In-Domain F1	Cross-Domain Acc	Cross-Domain F1
mean	0.3731	0.3470	0.2296	0.1317
max	0.3769	0.3587	0.2741	0.2063
mean_std	0.4500	0.4368	0.2096	0.1536

6.3 Comparative Summary

- **Best in-domain**: wav2vec2 + max (Macro-F1 = 0.5852)
- **Best cross-domain**: MFCC + max (Macro-F1 = 0.2063)
- **Best pooling overall** (average across both models and both settings): mean_std

7 Discussion

7.1 In-Domain vs Cross-Domain Gap

Both feature families show a clear drop from in-domain to cross-domain performance, confirming significant domain shift between RAVDESS and CREMA-D. This is expected due to differences in speaker population, recording setup, style, and dataset bias.

7.2 Feature Family Behavior

wav2vec2 is clearly stronger in-domain, suggesting richer representations for within-domain separability. However, in the current cross-domain setup, MFCC+max generalized better than wav2vec2 variants, indicating that robust low-level summaries can sometimes transfer better without domain adaptation.

7.3 Pooling Effects

Pooling strongly affects results:

- `max` is strongest for cross-domain in both families.
- `mean_std` gives the strongest overall aggregate balance.

8 What Is Implemeted

- Modular preprocessing and dataset integration (RAVDESS + CREMA-D)
- Unified label mapping for cross-dataset evaluation
- MFCC and wav2vec2 pipelines
- Speaker-independent in-domain and cross-domain experiments
- Pooling study
- Class balancing in classifier
- Reproducible logging/export artifacts

9 Conclusion

The implemented system satisfies the intended two-dataset SER research workflow with submission-quality reproducibility: robust preprocessing, reusable feature extraction, resumable caching, controlled splits, and complete result export. The experiments establish both a strong in-domain reference and a realistic cross-domain generalization benchmark. Future work should prioritize third-dataset integration (IEMOCAP) and domain adaptation methods to improve transfer performance.